Nikita Kazeev

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Research Scientist at the intersection of AI and Physics

Work Experience

National University of Singapore *Postdoc under Kostya Novoselov* **2022–present**

Wyckoff Transformer for generating symmetric crystals ICML 2025 · code

- Developed a coordinate-free permutation-invariant autoregressive Transformer encoder model for material generation and property prediction based on symmetry inductive bias
- Increased materials discovery yield (S.U.N.) by 1.24x
- Accelerated inference throughput by 4 orders of magnitude, dropping generation costs to 0.05 GPU ms per structure.
- Implemented the model in PyTorch; production-ready package published on [PyPI](https://pypi.org/project/w transformer/
- Orchestrated large-scale (10k structures) DFT computations with VASP at an HPC cluster
- Led a team of 6

ML for predicting properties of defects in 2D crystals paper 1 · paper 2 · patent · code

- Conceived a novel sparse representation of crystals with defects
- Achieved a 3.7x energy prediction error reduction
- Optimized training and inference workflows, reducing memory usage by 4x and GPU operations by 8x
- Developed code for reproducible parallel running of machine learning experiments
- Led a team of 3

Constructor Group *Machine Learning Consultant* **2021–present**

CERN [Yandex → HSE University] *Researcher* **2014–2022**

2025 Breakthrough Prize in Fundamental Physics as a member of LHCb.

Generative models uncertainty estimation paper · conference · code 1 · code 2

- Developed the first methods for estimating uncertainty of conditional GANs.
- Led a team of 3 students.

Machine learning on noisy data paper 1 · paper 2 · code

- Developed a rigorous way to train ML algorithms on data with the label-noise model common in high-energy physics.

Muon identification at the LHCb experiment at CERN paper · poster · IDAO-2019

- Solved a classification problem over tabular data with noisy labels under timing constraints.
- Developed a gradient boosting model that reduced the false positive rate by 30% in the critical low-track-momentum region.
- Integrated the model into LHCb production, mostly in C++.
- Packaged the work as a data science competition problem for IDAO-2019.

CatBoost aka fighting biases with dynamic boosting paper

- Set up a distributed system for running experiments for the team with Bash and Python.
- Studied gradient boosting improvements with experiments on toy data.

EDUCATION

Higher School of Economics (HSE University) **2016–2020**
PhD in Computer Science, supervisor Andrey Ustyuzhanin; Machine Learning for particle identification in the LHCb detector.

Sapienza — Universita di Roma **2016–2020**
PhD in Physics (double degree with HSE), supervisor Barbara Sciascia.

Product Management course at Yandex **2018**
Focused coursework in product strategy and execution.

Yandex School of Data Analysis **2013–2015**
Master's level CS course covering algorithms, machine learning, deep learning, and distributed systems.

Moscow Institute of Physics and Technology
MS in Physics (2014–2016): Optimisation of data processing of the LHCb experiment.
BS in Physics (2010–2014): Study of the quantum states of the electrons in nonideal plasma and selected molecules using wave packet molecular dynamics with packet splitting.

MENTORSHIP

- Mentored 9 students (1, 2, 3, 4, 5, 6, 7, 8, 9), 3 interns, and student workshop projects (1, 2).

TEACHING & OUTREACH

- Taught at the Machine Learning in High Energy Physics Summer School in 2015, 2016, 2017, 2018, 2019, 2020, 2021, and 2023.
- Lecturer in the Coursera Advanced Machine Learning specialisation, which is award-winning.
- Taught and developed an introductory data analysis course at HSE.
- Delivered more than 20 conference talks.

SERVICE

- Reviewer for RSC Advances, Machine Learning: Science and Technology, and AI for Accelerated Materials Design workshops at NeurIPS and ICLR (2023–2025).
- Peer Staff Supporter at NUS, serving as a first-line support for mental wellbeing.

OTHER

- Aspiring stand-up comedian; bombed more than 20 times at open mics.
- International Junior Science Olympiad 2008 in Korea, silver medal.

Skills

- **ML & AI** — Structured, non-text, domain-specific data
- **Python** — Research coding
- **PyTorch** — Deep learning
- **C++** — Production
- **Linux** — Administration
- **Physics** — Deep domain understanding

Selected Publications

1. **Kazeev, Nikita**, et al. WyckoffTransformer: Generation of Symmetric Crystals. ICML 2025, May 2025.
2. **Kazeev, N.**, Al-Maeeni, A.R., Romanov, I. et al. Sparse representation for machine learning the properties of defects in 2D materials. npj Comput Mater 9, 113 (2023).
3. M. Borisyak and **N. Kazeev**. Machine Learning on data with sPlot background subtraction. Journal of Instrumentation 14.08 (2019). **Alphabetic order, I'm the corresponding author.**
4. Derkach, D., **Kazeev, N.**, Ratnikov, F., Ustyuzhanin, A., & Volokhova, A. (2019). Cherenkov detectors fast simulation using neural networks. *Nuclear Instruments and Methods in Physics Research Section A*. **Alphabetic order, I'm the corresponding author.**